Rocket Designing

1. Mission target & requirements

Target: Max. vertical separation btw booster and dart apogees
Payload: Prepare a mounting location for the competition flight recorder
Req:

An avionics system to record Altitude & Acceleration

GPS for dart & optional for booster

A Video recording device in dart: down-looking

labeling: CP CG @ full-loaded & @burnout

Thrust: motor provided, to weight ratio >5:1

Altitude: No exceed 4k AGL

Recovery: dual-deployment for dart (Drogue and Main), single parachute in booster

Dart & Booster in a ready-to-fly condition

2. Timeline & Flight sequence of events

Pre-flight (T-Minus)

Battery charging, Flight-computer standby, Altimeter setting ...

T-Plus (Autonomous Flight Sequence)

Time	Event	Purpose
T + 0.00 s	Liftoff, Rod clearance	The rocket clears the launch rod, start climbing
T + 0.70 s	Mach 1	Breaks the sound barrier, 767 mph, 400 ft AGL
T + 1.10 s	Motor burnout, Max-Q, Booster separate	900 ft, ~ 1000mph Mach 1.3
T + 4.00 s	Booster Apogee	~ 1300 ft, Dart climbing through 3500 ft
T +15.00 s	Dart Apogee	~ 6000 ft (we have 4000 ft AGL ceiling limits)

Recovery

T + 15.01 s	Dart Drogue deployment	
T + 17.00 s	Booster parachute deploy	Booster descending passes 500 ft, speed 60 fps
T + 35.00 s	Booster touchdown	Lands safely at 17 fps
T + 95.00 s	Dart main parachute deployment	Descend passses 500 ft, speed 100 fps
T +125.00 s	Dart touchdown	Lands safely at 16 fps

3. Pickup a motor

// 1. the “container” law >> motor dictates the structural geometry

2. Mandatory baseline, other discipline relies on motor’s physical and performance constants

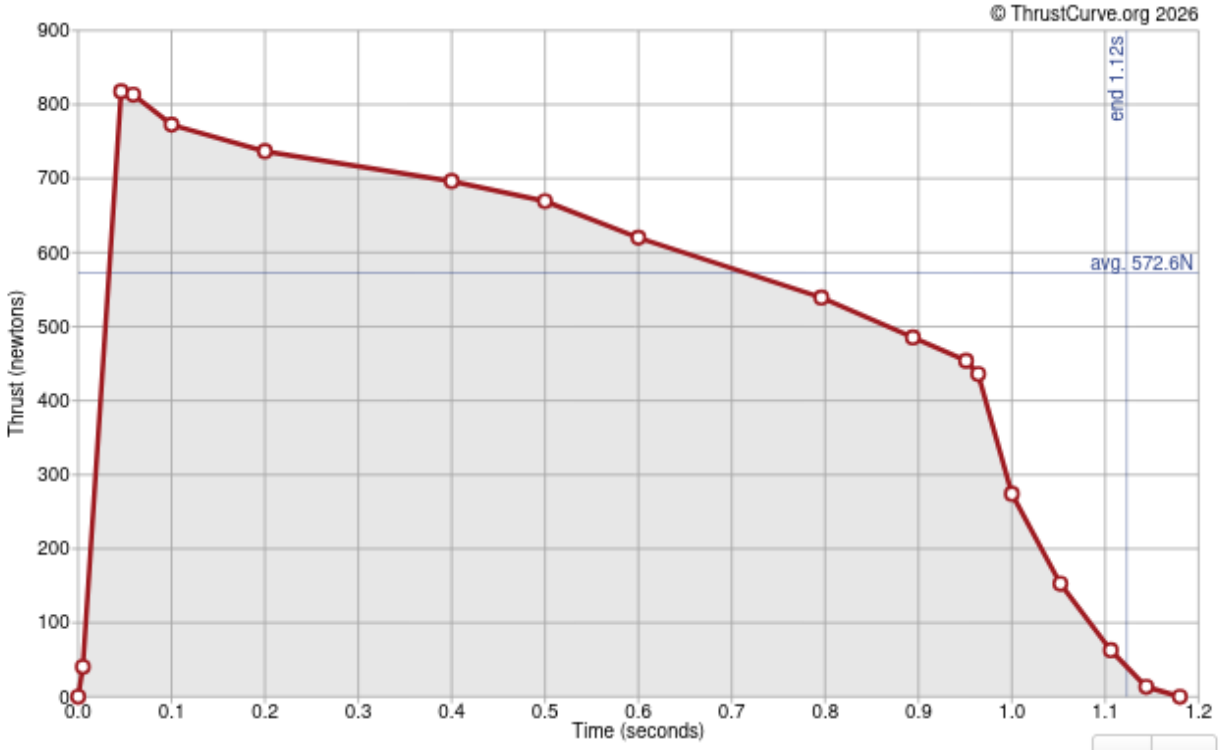
3. Moving CG, propellant burnout but empty (unburned) weight pull the CG down

4. motor controls the lift-off physics

e.g. Rail exit velocity. Motor must build up thrust quickly to accelerate the rocket “fast enough” for safely leaving the launch rod

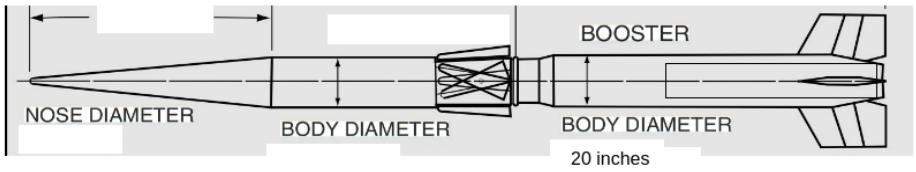
Now, engine team to provides motor data

AeroTech RMS 38mm I600 Redline
Dimension: 38 mm(1.5 in) diameter x 345 mm(13.6 in) length
Weight: 605 g total, 324g propellant
Thrust: 696 N Initial, 612 N Average, 812 N Peak
Impulse: 597 ~ 640 Ns total



4. Spatial Layout

for **team leader**/ Integration lead / Chief Engineer >> the ultimate Integrator
is the battleground of subsystem trade-offs
CG control
…



example from Jing (discussion only)

booster:

40 mm outer diameter, 1 mm thickness, 20 inches (508 mm) length, 6061 **aluminum tube**: 170g
3 x fins, 80mm root, 35mm tip, 50mm span, 3 x 12g = 36 g
to mount the motor, needs a 38mm **aft motor retainer** ~ 10 g
38mm solid disk with thicknesss of ? mm bulkhead ~20 g
240 g empty

Parachute: 80 g
30 inches flat-round nylon, drag coefficient ~0.8, descent speed <20ft/s, ~40 g + 40g shield&cord

Dart:

Component	Assumptions	Weight
Fiberglass Airframe	16 inches tube, 1in OD	100 g
Nose cone	5:1 Ogive profile, PEGT 3D print	50 g
Fins	X 3, fiberglass swept profile	30 g
Dual recovery system		
Drogue	In nose cone + forward section, 3 in space	20 g
Main	Forward section 8 inches space, 12 inches thin nylon	20 g
Avionics:	In center section	60g
Payload bay	Aft section	
Camera	Attached @ fin root	10g
Internal support structue		50 g
	Total:	340 g

Rocket Empty weight: 660 g
Motor: 605 g
payload (competition flight recorder) ~50g

lift-off weight: 1305 g

5. sub-system kick in

Structural:
- material selection
- wall thickness
- separation coupler
- nose cone, fin attachment, ...

digital model if 3D printing, numerical static analysis, CG data…

Aerodynamic:
- Fin design:
- CP
- drag minimizing

Electronic:
- Raspberry Pi

Recovery:
- booster parachute: size, opening shock, landing touchdown speed
- Dart dual system

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6. Iterating

layout >> structure >> CG >> aerodynamic CP >> Stability
>> electronic >> trajectory and simulation
>> recovery
if not working, back to layout

7. write a report, buy materials, fabricate, test, launch